

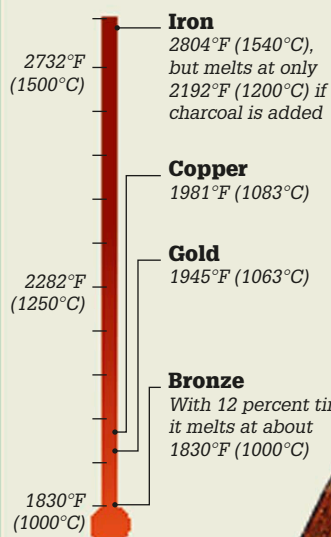


Bull-shaped gold ornament from a burial site in Varna, Bulgaria

Earliest metalworking

Some metals, especially copper and gold, can occur naturally as nuggets. Around 9000 BCE, metalworkers discovered that hammering such metals into thin sheets made them hard enough to fashion into simple objects, such as ornaments.

Heat needed to melt metals



Metalworking

From around 9000 BCE, people began to use naturally occurring metal for making tools instead of stone, bone, or wood. Then, craftsmen discovered how to melt out metal from metal-bearing rocks by using intense heat. First they worked with copper, then bronze (a mix, or alloy, of copper and tin), and finally, iron. As technology advanced, tools and weapons became stronger and more durable than before.

Bronze axhead with human mask design, Shang Dynasty (12th–11th century BCE), China



Smelting copper

By about 5500 BCE, people were extracting copper from its ore (rock in which a metal is embedded) by a process called smelting. This involved heating copper-bearing rocks to high temperatures in a furnace. The molten copper ran off and was molded or beaten into shape while cooling.

Discovery of bronze

Metalworkers discovered that adding another metal to copper while it was at a high temperature produced bronze. An alloy, bronze is harder than the original metals. At first, from around 4200 BCE, bronze was made by adding arsenic to copper. Then from 3200 BCE, metalworkers used a mixture containing 12 percent tin.



Egyptian metalworkers heating copper

Crucible contains copper ore that is heated until it melts and releases copper.

Key events

c 9000 BCE

Cold-working of copper and gold, by beating or hammering the pure metals into thin strips or sheets, was developed in the Balkans, in southeastern Europe.

c 5500 BCE

Smelting of copper was discovered in the Balkans and Anatolia. It spread rapidly through the Middle East and to Egypt.

4200 BCE

Arsenic was added to copper during smelting to produce a form of bronze.

3200 BCE

Tin was added to copper to produce tin bronze, which is harder than copper, and could be used to make better arms and armor.

c 2500 BCE

Early iron production created a metal that was soft and easily shaped, but did not produce strong objects.